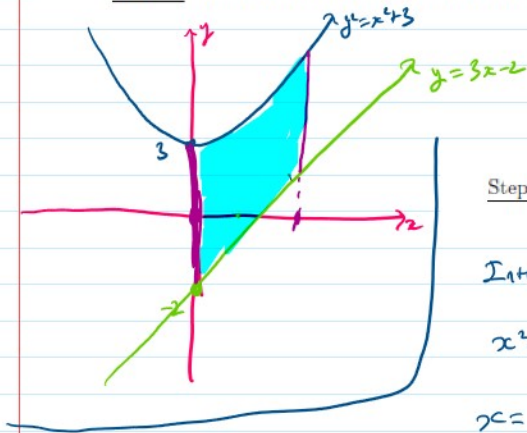


Area Between Two Curves

Example 1. Find the area between the curves $y = x^2 + 3$ and $y = 3x - 2$ on the interval $[0, 1]$.

Step 1: Sketch the two curves and shade the region whose area we are trying to find.



Step 2: Are there any intersection points within the interval $[0, 1]$?

Intersection happens when $x^2 + 3 = 3x - 2$

$$x^2 - 3x + 5 = 0 \quad \left(x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \text{ for } ax^2 + bx + c = 0 \right)$$

$$x = \frac{-(-3) \pm \sqrt{(-3)^2 - 4 \cdot 1 \cdot 5}}{2 \cdot 1}$$

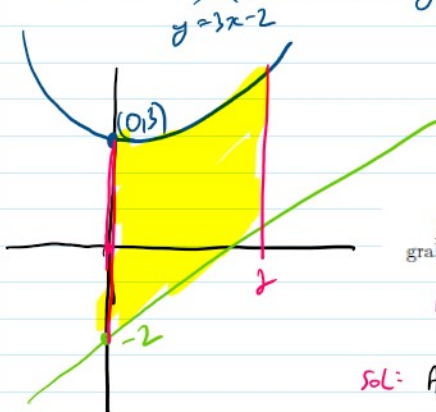
$$= \frac{3 \pm \sqrt{9 - 20}}{2} = \frac{3 \pm \sqrt{-11}}{2}$$

$\Rightarrow$ There is no real roots, so there is no intersection.

Step 3: Which curve lies above the other?

Pick a point $\Rightarrow x = 0$. At $x = 0$, first curve $y = x^2 + 3$ passes through $(0, 0+3) = (0, 3)$

The second curve passes through $(0, 3 \cdot 0 - 2) = (0, -2)$



As $-2 < 3$, the curve $x^2 + 3$ is above than the curve $y = 3x - 2$.

Step 4: Calculate the area of the region by setting up and solving the appropriate integral.

A) 4 B) $\frac{11}{3}$ C) $\frac{7}{2}$ D) $\frac{23}{6}$ E) None.

$$\text{Sol: Area} = \int_0^1 (x^2 + 3) - (3x - 2) dx = \int_0^1 x^2 - 3x + 5 dx =$$

$$= \left. \frac{x^3}{3} - \frac{3x^2}{2} + 5x \right|_{x=0}^{x=1} = \frac{1^3}{3} - \frac{3 \cdot 1^2}{2} + 5 - 0 = \frac{23}{6}$$

Example 2. Find the area between the curves $y = x\sqrt{10-x^2}$ and $y = x$.

Step 1: Where are the intersection points?

Hint: Do it analytically. Do not try to draw.

Then, solve x for $x\sqrt{10-x^2} = x$.

Note:

Off: Tuesdays 12³⁰-1³⁰ pm

Then, solve x for $x\sqrt{10-x^2} = x$.

$$x\sqrt{10-x^2} - x = 0 \quad x(\sqrt{10-x^2} - 1) = 0$$

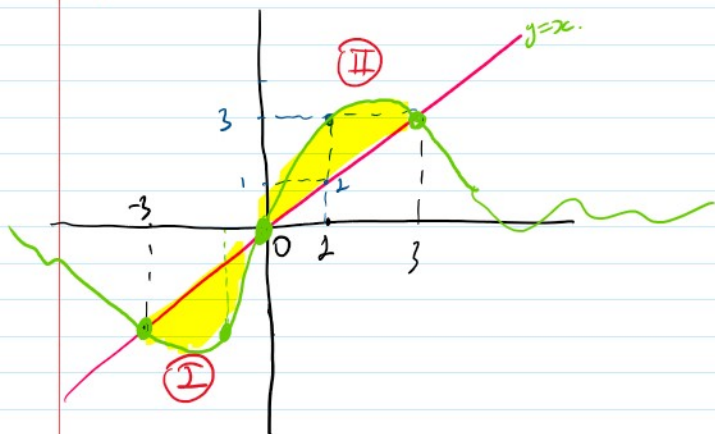
$$x=0 \text{ or } \sqrt{10-x^2} = 1$$

$$10-x^2 = 1$$

$$x^2 = 9 \quad x = \pm 3$$

3 intersection points: $-3, 0, 3$

Step 2: When $0 < x < 3$, which curve lies above the other? What about when $-3 < x < 0$?



Pick $x=1$: $y = x\sqrt{10-x^2}$
 $= 1 \cdot \sqrt{10-1^2} = 3$ when $x=1$.

So, I know $y = x\sqrt{10-x^2}$ passes through
 $(x, y) = (1, 3)$

Pick $x=-1 \Rightarrow y = -1 \cdot \sqrt{10-(-1)^2} = -3$

Similarly, $y = x\sqrt{10-x^2}$ passes through $(-1, -3)$

Step 3: Calculate the area of the region by setting up and solving the appropriate integral.

Answer: $\frac{a \cdot b^{\frac{3}{2}} - c}{d}$

A) $\frac{2 \cdot 7^{\frac{3}{2}} - 11}{4}$ B) $\frac{10 \cdot 10^{\frac{3}{2}} - 3}{5}$

C) $\frac{2 \cdot 10^{\frac{3}{2}} - 29}{3}$ D) $\frac{3 \cdot 10^{\frac{3}{2}} - 13}{3}$ E) None.

Sol: $A = \int_{-3}^0 x - x\sqrt{10-x^2} dx + \int_0^3 x\sqrt{10-x^2} - x dx$

I II

I = $\frac{x^2}{2} \Big|_{x=-3}^0 - \int_{-3}^0 x\sqrt{10-x^2} dx$

$-\frac{9}{2}$

So, $u = 10 - x^2$ $u = 1$ if $x = -3$
 $u = 10$ if $x = 0$

$du = -2x dx$ $x dx = -\frac{du}{2}$

$\int_1^{10} \sqrt{10-x^2} x dx = \int_1^{10} \sqrt{u} \cdot -\frac{du}{2} = -\frac{1}{2} \int_1^{10} \sqrt{u} du =$

$$= -\frac{1}{2} \frac{2}{3} u^{\frac{3}{2}} \Big|_{u=1}^{10} = \frac{-u^{\frac{3}{2}}}{3} \Big|_{u=1}^{10} =$$

$$= -\frac{(10)^{\frac{3}{2}}}{3} + \frac{1}{3}$$

$$\textcircled{I} = -\frac{9}{2} - \left(-\frac{(10)^{\frac{3}{2}}}{3} + \frac{1}{3} \right) = \frac{(10)^{\frac{3}{2}}}{3} - \frac{1}{3} - \frac{9}{2} = \frac{10^{\frac{3}{2}}}{3} - \frac{1}{3} - \frac{9}{2}$$

Similarly, we calculate $\textcircled{II} = -\frac{9}{2} - \frac{1}{3} + \frac{10^{\frac{3}{2}}}{3}$

The answer will be (C).